

Q1 (3)

$$e^{(\cos^2 x + \cos^4 x + \dots \infty) \ln 2} = 2^{\cos^2 x + \cos^4 x + \dots \infty}$$

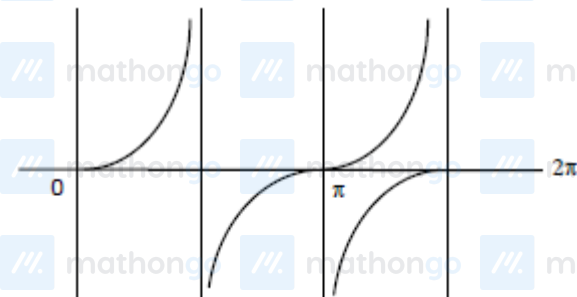
$$= 2^{\cot^2 x}$$

$$t^2 - 9t + 8 = 0 \Rightarrow t = 1, 8$$

$$\Rightarrow 2^{\cot^2 x} = 1, 8 \Rightarrow \cot^2 x = 0, 3$$

$$50 \Rightarrow \frac{2 \sin x}{\sin x + \sqrt{3} \cos x} = \frac{2}{1 + \sqrt{3} \cot x} = \frac{2}{4} = \frac{1}{2}$$

Q2 (2)



$$\tan 2\theta(1 + \cos 2\theta) > 0$$

$$2\theta \in \left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right) \cup \left(2\pi, \frac{5\pi}{2}\right) \cup \left(3\pi, \frac{7\pi}{2}\right)$$

$$\Rightarrow \theta \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{5\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{7\pi}{4}\right)$$

Q3 (1)

$$2 \cos\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) - \left[2 \cos^2\left(\frac{x+y}{2}\right) - 1\right] = \frac{3}{2}$$

$$2 \cos\left(\frac{x+y}{2}\right) \left[\cos\left(\frac{x-y}{2}\right) - \cos\left(\frac{x+y}{2}\right)\right] = \frac{1}{2}$$

$$2 \cos\left(\frac{x+y}{2}\right) \left[2 \sin\left(\frac{x}{2}\right) \cdot \sin\left(\frac{y}{2}\right)\right] = \frac{1}{2}$$

$$\cos\left(\frac{x+y}{2}\right) \cdot \sin\left(\frac{x}{2}\right) \cdot \sin\left(\frac{y}{2}\right) = \frac{1}{8}$$

$$\text{Possible when } \frac{x}{2} = 30^\circ \& \frac{y}{2} = 30^\circ$$

$$x = y = 60^\circ$$

$$\sin x + \cos y = \frac{\sqrt{3}}{2} + \frac{1}{2} = \frac{\sqrt{3}+1}{2}$$

Q4 (11)

Hints and Solutions

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$$3 \sin x + 4 \cos x = k + 1$$

$$-5 \leq k + 1 \leq 5$$

$$-6 \leq k \leq 4$$

$$-6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4 \mid \Rightarrow 11 \text{ integral values}$$

Q5 (1)

$$\sqrt{3}t^2 - (\sqrt{3} - 1)t - 1 = 0 \text{ (where } t = \cos x \text{)}$$

$$\text{Now, } t = \frac{(\sqrt{3}-1) \pm \sqrt{4+2\sqrt{3}}}{2\sqrt{3}}$$

$$t = \cos x = 1 \text{ or } -\frac{1}{\sqrt{3}} \rightarrow \text{rejected as } x \in \left[0, \frac{\pi}{2}\right]$$

$$\Rightarrow \cos x = 1$$

$$\Rightarrow \text{No. of solution} = 1$$

Questions with Answer Keys

MathonGo

Q1: 24 Feb (Shift 1) - Single Correct

If $e^{(\cos^2 x + \cos^4 x + \cos^6 x + \dots) \log_e 2}$ satisfies the equation $t^2 - 9t + 8 = 0$, then the value of

$\frac{2 \sin x}{\sin x + \sqrt{3} \cos x} \left(0 < x < \frac{\pi}{2}\right)$ is

(1) $\frac{3}{2}$

(2) $2\sqrt{3}$

(3) $\frac{1}{2}$

(4) $\sqrt{3}$

Q2: 25 Feb (Shift 1) - Single Correct

All possible values of $\theta \in [0, 2\pi]$ for which $\sin 2\theta + \tan 2\theta > 0$ lie in:

(1) $\left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right)$

(2) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{5\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{7\pi}{4}\right)$

(3) $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{7\pi}{6}\right)$

(4) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{11\pi}{6}\right)$

Q3: 25 Feb (Shift 2) - Single Correct

If $\sin \theta = \frac{1}{2}$

(1) $\frac{1+\sqrt{3}}{2}$

(2) $\frac{1-\sqrt{3}}{2}$

(3) $\frac{\sqrt{3}}{2}$

(4) $\frac{1}{2}$

Q4: 26 Feb (Shift 1) - Numerical

The number of integral values of 'k' for which the equation $3 \sin x + 4 \cos x = k + 1$ has a solution, $k \in R$ is

Q5: 26 Feb (Shift 1) - Numerical

If $\sqrt{3} (\cos^2 x) = (\sqrt{3} - 1) \cos x + 1$, the number of solutions of the given equation when $x \in \left[0, \frac{\pi}{2}\right]$ is

Answer Key**Q1 (3)****Q2 (2)****Q3 (1)****Q4 (11)****Q5 (1)**